



AIKnowledgeassistant
.txt



AI_QA_Knowledge_Assistant_Ingestion_Arch



AI_QA_Knowledge_Assistant_Ingestion_Arch

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**I — Instruction**
I want to build an AI QA Knowledge Assistant application from scratch that can help any team to get the response specific to application on the basis of existing application information
Create a complete phase-by-phase technical architecture and implementation guide for ingestion that I can follow to build the application step by step.
[STRICT]: DO NOT HALLUCINATE
[STRICT]: Do not invent requirements or test cases.
[STRICT]: Distinguish documented facts from AI suggestions
[STRICT]: If evidence is insufficient, say "Insufficient evidence".
[STRICT]: Cite the source document for every important conclusion.
[STRICT]: Fetch all the values from .env file. DO NOT HARD CODE THE VALUES (example DB name, Connection String)
[STRICT]: Create one example.env

**C — Context**
The application should use the below input sources and process the same .
V stands for vector embedding(Those inputs to be converted as vector)
example :Requirements[Userstory//BRD,FRD,.etc](V)

*Input sources:
Requirements[Userstory//BRD,FRD,.etc](V)
Figma,Swagger API,UI specs(V)
Existing testcases(V)
testdata(V)
Jira/ADO/Testrail/Jysher//x-ray(V)
Developer Code repo(Github/bitbucket)(No Vector)
Meeting notes/recording(V)
Release notes (V)
Confluence //wiki //technical documentation(v)
Defect database(v)
Release Notes

The information related to Requirements, test cases and JIRA defects should extract useful information, generate embeddings, and store the processed requirements, test case, defect data in MongoDB.

**System Design
Functionalities(Critical+optional)

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**System Design
Functionalities(Critical+optional)
Critical :
Ingestion pipeline—It will help you to ingest all the data in vector DB
Step process:
Input->Embedding—Store in Mongodb
Final response for the user query generated by LLM

**S — Scope**
Cover the complete backend ingestion, retrieval and front end journey starting from project setup and ending with a fully stored and searchable query.
The architecture should cover areas such as:
- Project setup
- Folder structure
- Environment configuration
- Express server
- MongoDB connection
- Document PDF/docs/excel/notepad upload
- File validation
- PDF text extraction
- Text cleaning
- Documents parsing
- Regex utilities
- Skills identification
- Structured metadata extraction
- Optional LLM parser
- Mistral embedding generation
- MongoDB storage
- Complete ingestion orchestration
- Error handling
- Logging
- Testing
- Final verification

Design the folder structure so that a Ingestion,Retrieval and UI module can be added to the same codebase without restructuring the entire application.

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Design the folder structure so that a Ingestion,Retrieval and UI module can be added to the same codebase without restructuring the entire application.

Technology preference:
- Node.js
- TypeScript
- React.js
- Express
- MongoDB
- Mistral Embeddings
- PDF resume processing/Word Doc processing/Excel processing/url processing
- Algorithm/Regex-based resume parsing
- Optional LLM-based parsing
- LLM -Grog //openai
-Embedding model-Mistral //openai

Deployment server :
AWS

**E — Expectation**
Do not jump directly to the complete application. Break the implementation into small, logical phases.
For every phase, clearly explain:
- What we are building
- Why that phase is required
- Which folders/files need to be created
- What logic needs to be implemented
- Which API needs to be created, if applicable
- How I can run and test that phase
- What I should verify before moving forward
- What failure/error scenarios should be handled

Each phase must be independently testable.
I should move to the next phase only after the current phase is successfully implemented and verified.

**E - Example
```

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**E - Example
1. Test Coverage suggestion
User asks:"We are changing the customer address validation API. What should I test based on previous defects and existing test cases?"
The answer could look like:
Area Recommendation Source
Mandatory fields Verify missing/blank address fields TC-102
Boundary Test maximum address length DEF-231
Special characters Validate characters rejected previously DEF-417
API response Verify 400/422 responses API Spec
Regression Re-test customer profile update Release-24.6

**E - Example
2. Requirement → Existing Test Cases
User asks:"Show me test cases related to customer address validation."
RAG retrieves relevant cases.
Output:
Requirement
↓
Relevant test cases
↓
Coverage status
↓
Missing areas

**Example
3. Defect Similarity Search
User: "Find previous defects similar to this issue."
Input:Customer address API returns 500 when apartment number contains special characters.
RAG searches historical defects and returns:
Similar Defect: DEF-417
Similarity: High

Previous behavior:
API failed for special characters in addressLine2.
Root cause:Input validation was missing
```

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Previous behavior:
API failed for special characters in addressLine2.
Root cause:Input validation was missing.
Resolution:Validation added before persis

**Example :
4. Requirement → Test Coverage Gap
User: "Is this requirement adequately covered by our existing test cases?"
The system retrieves:
Requirement
→ relevant test cases
→ historical defects
→ acceptance criteria
Then produces: Coverage Found

Functional      ✓
Positive       ✓
Negative       ✓
Boundary      X
API validation ✓
Error handling X

Potential gaps:
1. Maximum address length
2. Special characters
3. Null addressLine2
Notice that this is not simply document Q&A.

**Example:
5. Defect → Regression Impact
User: "DEF-417 was fixed. What should be included in regression?"
RAG retrieves:
DEF-417
↓
Root cause
↓
Affected requirement
↓
Related test cases
↓
Related historical defects
↓
Regression candidates
Output:
Recommended regression areas:

Customer Profile
├─ Address creation
├─ Address modification
├─ Address search
├─ Address validation API

Reason:
These areas were linked to the affected requirement in
historical test/defect records.

Example :
6. QA Question Answering
A tester can simply ask:
"What is the expected behavior when addressLine2 is blank?"
The assistant searches the organization's actual requirements and gives:
Expected behavior:
AddressLine2 is optional.
```

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**Example:
5. Defect → Regression Impact
User: "DEF-417 was fixed. What should be included in regression?"
RAG retrieves:
DEF-417
↓
Root cause
↓
Affected requirement
↓
Related test cases
↓
Related historical defects
↓
Regression candidates
Output:
Recommended regression areas:

Customer Profile
├─ Address creation
├─ Address modification
├─ Address search
├─ Address validation API

Reason:
These areas were linked to the affected requirement in
historical test/defect records.

Example :
6. QA Question Answering
A tester can simply ask:
"What is the expected behavior when addressLine2 is blank?"
The assistant searches the organization's actual requirements and gives:
Expected behavior:
AddressLine2 is optional.
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Example :
6. QA Question Answering
A tester can simply ask:
"What is the expected behavior when addressLine2 is blank?"
The assistant searches the organization's actual requirements and gives:
Expected behavior:
AddressLine2 is optional.

Source:
Customer Address FRD v2.3
Section 4.2

Related test cases:
TC-231
TC-232
TC-245

****P — Persona****

Act as a Senior Architect and RAG Application Architect who is guiding a developer building this application for the first time.

Assume I understand basic development concepts but need clear architectural guidance and implementation sequencing.

****O — Output****

Generate one complete Markdown architecture document.

Organize it phase by phase for Ingestion only.

The document should be detailed enough that I can give it to a developer or AI coding assistant and build the backend by following the phases sequentially.

Clearly include verification gates between phases and finish with a final ingestion readiness checklist.

The final architecture must clearly establish:

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TC-231
TC-232
TC-245

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The final architecture must clearly establish:

**** Ingestion(from any source including pdf, doc, JIRA, Confluence, excel) -> Verification -> Ingestion Complete****
Refer the put sources section in this prompt for type of input source

****T — Tone****

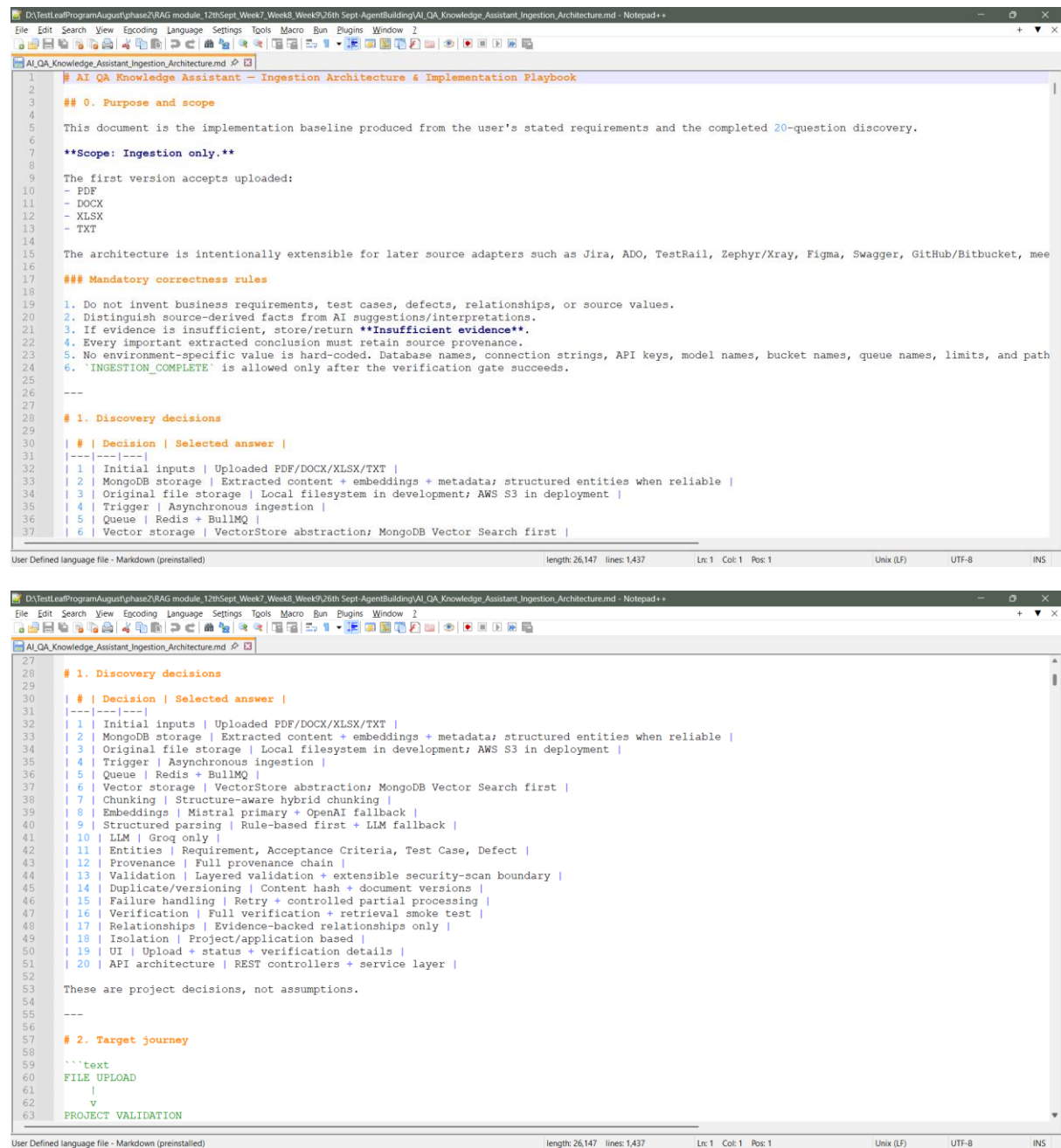
Use clear, practical, developer-friendly technical English.

Avoid unnecessary theory.

Write it like an implementation playbook that a first-time developer can follow from an empty project to a working AI knowledge Assistant.

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Architecture file generated



```
1  # AI QA Knowledge Assistant - Ingestion Architecture & Implementation Playbook
2
3  ## 0. Purpose and scope
4
5  This document is the implementation baseline produced from the user's stated requirements and the completed 20-question discovery.
6
7  **Scope: Ingestion only.**
8
9  The first version accepts uploaded:
10 - PDF
11 - DOCX
12 - XLSX
13 - TXT
14
15 The architecture is intentionally extensible for later source adapters such as Jira, ADO, TestRail, Zephyr/Xray, Figma, Swagger, GitHub/Bitbucket, mee
16
17 ### Mandatory correctness rules
18
19 1. Do not invent business requirements, test cases, defects, relationships, or source values.
20 2. Distinguish source-derived facts from AI suggestions/interpretations.
21 3. If evidence is insufficient, store/return **Insufficient evidence**.
22 4. Every important extracted conclusion must retain source provenance.
23 5. No environment-specific value is hard-coded. Database names, connection strings, API keys, model names, bucket names, queue names, limits, and path
24 6. 'INGESTION_COMPLETE' is allowed only after the verification gate succeeds.
25
26 ---
27
28 # 1. Discovery decisions
29
30 | # | Decision | Selected answer |
31 |---|---|
32 | 1 | Initial inputs | Uploaded PDF/DOCX/XLSX/TXT |
33 | 2 | MongoDB storage | Extracted content + embeddings + metadata; structured entities when reliable |
34 | 3 | Original file storage | Local filesystem in development; AWS S3 in deployment |
35 | 4 | Trigger | Asynchronous ingestion |
36 | 5 | Queue | Redis + BullMQ |
37 | 6 | Vector storage | VectorStore abstraction; MongoDB Vector Search first |
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57 # 2. Target journey
58
59 ```text
60 FILE UPLOAD
61 |
62 v
63 PROJECT VALIDATION
```

```
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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
56
57 # 2. Target journey
58
59 ```text
60 FILE UPLOAD
61 |
62 v
63 PROJECT VALIDATION
64 |
65 v
66 FILE + CONTENT VALIDATION
67 |
68 v
69 HASH / DUPLICATE / VERSION CHECK
70 |
71 v
72 STORE ORIGINAL FILE
73 |
74 v
75 CREATE INGESTION JOB
76 |
77 v
78 REDIS + BULLMQ
79 |
80 v
81 INGESTION WORKER
82 |
83 +--> Extract PDF/DOCX/XLSX/TXT
84 |
85 +--> Clean / normalize
86 |
87 +--> Rule-based parsing
88 |
89 | +--> Requirement
90 | +--> Acceptance Criteria
91 | +--> Test Case
92 | +--> Defect

```

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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
81 INGESTION WORKER
82 |
83 +--> Extract PDF/DOCX/XLSX/TXT
84 |
85 +--> Clean / normalize
86 |
87 +--> Rule-based parsing
88 |
89 | +--> Requirement
90 | +--> Acceptance Criteria
91 | +--> Test Case
92 | +--> Defect
93 |
94 +--> Groq fallback when deterministic parsing is insufficient
95 |
96 +--> Validate evidence + provenance
97 |
98 +--> Structure-aware chunking
99 |
100 +--> Mistral embeddings
101 |
102 | +--> failure -> OpenAI fallback
103 |
104 +--> MongoDB persistence
105 |
106 +--> Verification
107 |
108 | +--> storage
109 | +--> extraction
110 | +--> entities
111 | +--> chunks
112 | +--> embeddings
113 | +--> provenance
114 | +--> retrieval smoke test
115 |
116 v
117 INGESTION COMPLETE

```

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```
115 |
116 | v
117 | INGESTION_COMPLETE
118 |
119 | ...
120 | BullMQ provides worker execution and retry mechanisms; its documentation states that workers process queued jobs and failed jobs can be retried. [Bull
121 |
122 | Mistral's embeddings API accepts text input and a model and returns numerical embedding vectors; its API also supports an optional output dimension wh
123 |
124 | MongoDB Vector Search supports semantic search over stored embeddings and supports filtering on indexed fields, which is relevant to project-level iso
125 |
126 | ---
127 |
128 | # 3. Logical architecture
129 |
130 | ```text
131 | React UI
132 | |
133 | | v
134 | Express REST API
135 | |
136 | | +--> Project Service
137 | | +--> Upload Service
138 | | +--> Ingestion Job Service
139 | |
140 | | v
141 | BullMQ / Redis
142 | |
143 | | v
144 | Ingestion Worker
145 | |
146 | | +--> FileStorage abstraction
147 | | | +--> Local
148 | | | +--> S3
149 | | |
150 | | +--> Validation
151 | | +--> Extractors
```

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```
143 v
144 Ingestion Worker
145 |
146 +--> FileStorage abstraction
147 |   +--> Local
148 |   +--> S3
149 |
150 +--> Validation
151 +--> Extractors
152 |   +--> PDF
153 |   +--> DOCX
154 |   +--> XLSX
155 |   +--> TXT
156 |
157 +--> Cleaning
158 +--> Rule parsers
159 +--> Groq parser
160 +--> Provenance
161 +--> Chunking
162 +--> EmbeddingService
163 |   +--> Mistral
164 |   +--> OpenAI fallback
165 |
166 +--> Repository layer
167 |   +--> MongoDB
168 |   +--> VectorStore abstraction
169 |
170 +--> Verification
171 |
172 v
173 MongoDB
174 +--> documents
175 +--> versions
176 +--> jobs
177 +--> chunks
178 +--> requirements
179 +--> acceptance criteria
```

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AI_QA_Knowledge_Assistant_Ingestion_Architecture.md

```
173 MongoDB
174 +--> documents
175 +--> versions
176 +--> jobs
177 +--> chunks
178 +--> requirements
179 +--> acceptance_criteria
180 +--> test_cases
181 +--> defects
182 +--> relationships
183 +--> provenance
184 +--> verification_results
185
186 Original file:
187 Local filesystem in development
188 AWS S3 in deployment
189 ...
190
191 AWS's current JavaScript SDK documentation describes S3 access through the SDK and recommends configuring authentication rather than embedding credent
192
193 ---
194
195 # 4. Repository structure
196
197 ```text
198 ai-qa-knowledge-assistant/
199 |
200 +-- apps/
201 |   +-- api/
202 |     |   +-- src/
203 |     |   |   +-- server.ts
204 |     |   |   +-- app.ts
205 |     |   |   +-- config/
206 |     |   |   +-- routes/
207 |     |   |   +-- controllers/
208 |     |   |   +-- middleware/
209 |     |   |   +-- workers/
```

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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
228 |
229 | | +-- retrieval/
230 | | | +-- domain/
231 | | | +-- application/
232 | | | +-- infrastructure/
233 | | |
234 | | +-- knowledge/
235 | | | +-- domain/
236 | | | +-- repositories/
237 | |
238 | +-- packages/
239 | | +-- config/
240 | | +-- logging/
241 | | +-- shared/
242 | | +-- schemas/
243 | |
244 | +-- tests/
245 | | +-- unit/
246 | | +-- integration/
247 | | +-- contract/
248 | | +-- fixtures/
249 | | +-- e2e/
250 | |
251 | +-- docs/
252 | | +-- architecture/
253 | | +-- runbooks/
254 | |
255 | +-- .env
256 | +-- example.env
257 | +-- .gitignore
258 | +-- package.json
259 | +-- tsconfig.json
260 | +--
261 |
262 | The `modules/ingestion` module must not depend on UI code. Retrieval is a sibling module and can be implemented later without moving ingestion.
263 |
264 | ---
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AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
265 |
266 | # 5. Environment configuration
267 |
268 | Create `example.env`:
269 |
270 | ```env
271 | NODE_ENV=
272 | PORT=
273 |
274 | MONGODB_URI=
275 | MONGODB_DATABASE=
276 | MONGODB_VECTOR_INDEX_NAME=
277 |
278 | REDIS_URL=
279 | BULLMQ_QUEUE_NAME=
280 | BULLMQ_WORKER_CONCURRENCY=
281 | BULLMQ_MAX_ATTEMPTS=
282 | BULLMQ_BACKOFF_DELAY_MS=
283 |
284 | FILE_STORAGE_PROVIDER=
285 | LOCAL_STORAGE_ROOT=
286 | AWS_REGION=
287 | AWS_S3_BUCKET=
288 | AWS_S3_PREFIX=
289 |
290 | MAX_UPLOAD_BYTES=
291 | ALLOWED_EXTENSIONS=
292 | ALLOWED_MIME_TYPES=
293 |
294 | MISTRAL_API_KEY=
295 | MISTRAL_BASE_URL=
296 | MISTRAL_EMBEDDING_MODEL=
297 | MISTRAL_OUTPUT_DIMENSION=
298 |
299 | OPENAI_API_KEY=
300 | OPENAI_BASE_URL=
301 | OPENAI_EMBEDDING_MODEL=
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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
287 AWS_S3_BUCKET=
288 AWS_S3_PREFIX=
289
290 MAX_UPLOAD_BYTES=
291 ALLOWED_EXTENSIONS=
292 ALLOWED_MIME_TYPES=
293
294 MISTRAL_API_KEY=
295 MISTRAL_BASE_URL=
296 MISTRAL_EMBEDDING_MODEL=
297 MISTRAL_OUTPUT_DIMENSION=
298
299 OPENAI_API_KEY=
300 OPENAI_BASE_URL=
301 OPENAI_EMBEDDING_MODEL=
302
303 GROQ_API_KEY=
304 GROQ_BASE_URL=
305 GROQ_MODEL=
306
307 LLM_PARSER_ENABLED=
308 LLM_PARSER_MAX_INPUT_TOKENS=
309
310 EMBEDDING_TIMEOUT_MS=
311 LLM_TIMEOUT_MS=
312 EXTERNAL_API_MAX_RETRIES=
313
314 LOG_LEVEL=
315 LOG_FORMAT=
316
317 INGESTION_VERSION=
318 PARSER_VERSION=
319 CHUNKER_VERSION=
320 ...
321
322 Do not copy production secrets into `example.env`.
323
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AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
328 # 6. Core status model
329
330 ```text
331 UPLOADED
332 VALIDATING
333 VALIDATED
334 QUEUED
335 PROCESSING
336 EXTRACTING
337 PARSING
338 CHUNKING
339 EMBEDDING
340 PERSISTING
341 VERIFYING
342 INGESTION_COMPLETE
343 PARTIALLY_PROCESSED
344 FAILED
345 ...
346
347 Centralize state transitions. Services must not invent arbitrary states.
348
349 ---
350
351 # 7. Data model
352
353 ## Project
354
355 ```text
356 Project
357 - _id
358 - projectKey
359 - name
360 - status
361 - createdAt
362 - updatedAt
363 ...
364
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
365 ## Document
366
367 ```text
368 Document
369 - _id
370 - projectId
371 - originalFileName
372 - fileType
373 - mimeType
374 - contentHash
375 - storageProvider
376 - storageKey
377 - currentVersionId
378 - status
379 - createdAt
380 - updatedAt
381 ```
382
383 ## DocumentVersion
384
385 ```text
386 DocumentVersion
387 - _id
388 - documentId
389 - versionNumber
390 - contentHash
391 - sourceLocation
392 - parserVersion
393 - chunkerVersion
394 - ingestionVersion
395 - status
396 - createdAt
397 ```
398
399 ## IngestionJob
400
401 ```text
```

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```
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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
399 ## IngestionJob
400
401 ```text
402 IngestionJob
403 - _id
404 - projectId
405 - documentId
406 - documentVersionId
407 - bullJobId
408 - status
409 - currentStage
410 - progress
411 - attempt
412 - startedAt
413 - completedAt
414 - errorCode
415 - errorMessage
416 ```
417
418 ## Chunk
419
420 ```text
421 Chunk
422 - _id
423 - projectId
424 - documentId
425 - documentVersionId
426 - chunkIndex
427 - text
428 - section
429 - page
430 - sheet
431 - location
432 - entityType
433 - entityId
434 - embedding
435 - embeddingProvider
```

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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
435 - embeddingProvider
436 - embeddingModel
437 - embeddingDimension
438 - provenanceId
439 ...
440
441 ## Requirement
442
443 ```text
444 Requirement
445 - id
446 - projectId
447 - documentVersionId
448 - sourceChunkIds[]
449 - requirementId
450 - title
451 - description
452 - acceptanceCriteriaIds[]
453 - sourceEvidence[]
454 - extractionMethod
455 ...
456
457 'requirementId' is populated only when supported by source evidence.
458
459 ## Acceptance Criteria
460
461 ```text
462 AcceptanceCriteria
463 - id
464 - projectId
465 - documentVersionId
466 - sourceChunkIds[]
467 - requirementId
468 - text
469 - sourceEvidence[]
470 - extractionMethod
471 ...
472
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
```

```
D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
473 ## Test Case
474
475 ```text
476 TestCase
477 - id
478 - projectId
479 - documentVersionId
480 - sourceChunkIds[]
481 - testCaseId
482 - title
483 - preconditions
484 - steps
485 - expectedResult
486 - requirementIds[]
487 - sourceEvidence[]
488 - extractionMethod
489 ...
490
491 ## Defect
492
493 ```text
494 Defect
495 - id
496 - projectId
497 - documentVersionId
498 - sourceChunkIds[]
499 - defectId
500 - summary
501 - description
502 - rootCause
503 - resolution
504 - requirementIds[]
505 - testCaseIds[]
506 - sourceEvidence[]
507 - extractionMethod
508 ...
509
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
```

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D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
510 ## Relationship
511
512 ```text
513 Relationship
514 - _id
515 - projectId
516 - fromEntityType
517 - fromEntityId
518 - relationshipType
519 - toEntityType
520 - toEntityId
521 - evidenceChunkIds[]
522 - relationshipStatus
523 - extractionMethod
524 ```
525
526 Relationship status should distinguish:
527
528 ```text
529 DOCUMENTED
530 INSUFFICIENT_EVIDENCE
531 ```
532
533 Semantic similarity alone must not be persisted as a documented business relationship.
534
535 ---
536
537 # 8. Provenance
538
539 ```text
540 Provenance
541 - _id
542 - projectId
543 - documentId
544 - documentVersionId
545 - sourceFileName
546 - page
547
548 ---
549
550 length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
```

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D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
552 - sourceText
553 - sourceChunkId
554 - extractionMethod
555 - parserVersion
556 - extractedAt
557 ```
558
559 Every important extracted field should point to evidence.
560
561 If a reliable source location/value cannot be established:
562
563 ```text
564 evidenceStatus = INSUFFICIENT_EVIDENCE
565 ```
566
567 ---
568
569 # 9. Phase-by-phase implementation
570
571 ## Phase 0 - Bootstrap
572
573 ### Build
574
575 Create the repository structure, TypeScript configuration, configuration loader, logger, shared schemas, and test framework.
576
577 ### Files
578
579 ```text
580 package.json
581 tsconfig.json
582 .env
583 example.env
584 .gitignore
585
586 apps/api/src/server.ts
587 apps/api/src/app.ts
588
589 ---
590
591 length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
```

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D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
568
569 # 9. Phase-by-phase implementation
570 ## Phase 0 - Bootstrap
571 ### Build
572
573 Create the repository structure, TypeScript configuration, configuration loader, logger, shared schemas, and test framework.
574
575 ### Files
576
577 ```text
578 package.json
579 tsconfig.json
580 .env
581 example.env
582 .gitignore
583
584 apps/api/src/server.ts
585 apps/api/src/app.ts
586
587 packages/config/
588 packages/logging/
589 packages/shared/
590 packages/schemas/
591
592 modules/ingestion/
593 modules/retrieval/
594 modules/knowledge/
595
596 tests/
597 ```
598
599 ### API
600
601 `GET /health`
602
603
604
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
```

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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
601
602 ### API
603
604 `GET /health`
605
606 ### Run
607
608 ```text
609 npm install
610 npm run build
611 npm run test
612 npm run dev
613 ```
614
615 ### Gate
616
617 Proceed only when:
618
619 - build succeeds
620 - tests run
621 - API starts
622 - `/health` responds
623 - missing required environment configuration fails clearly
624
625 ### Failure scenarios
626
627 - missing `.env`
628 - invalid configuration
629 - unavailable port
630 - invalid type in environment variable
631
632 ---
633
634 ## Phase 1 - MongoDB + Redis
635
636 ### Build
637
638 Create MongoDB and Redis connection modules.
639
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
```

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D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
632
633 ## Phase 1 - MongoDB + Redis
634
635 ### Build
636
637 Create MongoDB and Redis connection modules.
638
639 ### Files
640
641 ```text
642 modules/knowledge/infrastructure/mongodb/
643 apps/api/src/infrastructure/redis/
644 ```
645
646 ### API
647
648 Extend `/health` to report:
649
650 ```text
651 api
652 mongodb
653 redis
654 ```
655
656 ### Gate
657
658 - MongoDB connects
659 - Redis connects
660 - unavailable dependency produces controlled health failure
661 - no secrets are returned
662
663 BullMQ requires Redis as its backend and its worker model supports asynchronous job processing. [BullMQ Quick Start](https://docs.bullmq.io/quick-start)
664
665 ---
666
667 ## Phase 2 - Project boundary
668
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
```

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D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
666
667 ## Phase 2 - Project boundary
668
669 ### Build
670
671 Project creation and lookup.
672
673 ### APIs
674
675 ```text
676 POST /api/projects
677 GET /api/projects/:projectId
678 ```
679
680 ### Gate
681
682 Create one project and retrieve it.
683
684 Reject ingestion requests without a valid project.
685
686 ---
687
688 ## Phase 3 - File storage abstraction
689
690 ### Build
691
692 ```text
693 FileStorage
694 +-- LocalFileStorage
695 +-- S3FileStorage
696 ```
697
698 Methods:
699
700 ```text
701 put()
702 get()
703 ```
704
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
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D:\Test\selfProgramAugust\phase2\RAG module_120Sept_Week7 Task8_Week9\26th Sept: AgentBuilding\AI_OA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++

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AI_OA_Knowledge_Assistant_Ingestion_Architecture.md

```
687
688 ## Phase 3 - File storage abstraction
689
690 ### Build
691
692 ```text
693 FileStorage
694   +-- LocalFileStorage
695   +-- S3FileStorage
696   ...
697
698 Methods:
699
700 ```text
701 put()
702 get()
703 delete()
704 exists()
705 getMetadata()
706   ...
707
708 Provider is selected through `.env`.
709
710 ### Gate
711
712 Upload a fixture and confirm the bytes are retrievable.
713
714 ---
715
716 ## Phase 4 - Upload + validation
717
718 ### API
719
720 ```text
721 POST /api/projects/:projectId/documents
722   ...
723
```

User Defined language file - Markdown (preinstalled)length: 26,147 lines: 1,437Ln: 1 Col: 1 Pos: 1Unix (LF)UTF-8INS

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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
751   ### Gate
752
753   No invalid file reaches the extraction worker.
754
755   ---
756
757   ## Phase 5 - BullMQ job orchestration
758
759   ### Build
760
761   ```text
762   IngestionQueue
763   IngestionWorker
764   ```
765
766   Job payload:
767
768   ```text
769   projectId
770   documentId
771   documentVersionId
772   ```
773
774   Do not place full file contents into the queue payload.
775
776   Configure retry count, backoff, and concurrency from `.env`.
777
778   ### Gate
779
780   Upload creates a job, worker consumes it, status is persisted, and transient failure retry works.
781
782   BullMQ documents automatic retries through job `attempts` and backoff strategies. [Retrying failing jobs](https://docs.bullmq.io/guide/retrying-failin
783
784   ---
785
786   ## Phase 6 - Extraction
787
User Defined language file - Markdown (preinstalled)    length: 26,147    lines: 1,437    Ln: 1    Col: 1    Pos: 1    Unix (LF)    UTF-8    INS
```

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D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
786   ## Phase 6 - Extraction
787
788   ### Build
789
790   ```text
791   DocumentExtractor
792   +-- PdfExtractor
793   +-- DocxExtractor
794   +-- XlsxExtractor
795   +-- TxtExtractor
796   ```
797
798   Common output:
799
800   ```text
801   ExtractedDocument
802   - rawText
803   - pages[]
804   - sections[]
805   - paragraphs[]
806   - tables[]
807   - sheets[]
808   ```
809
810   Do not invent page/section/cell locations that the source extractor cannot provide.
811
812   ### Gate
813
814   Each supported format produces a valid intermediate representation or a controlled extraction error.
815
816   ---
817
818   ## Phase 7 - Cleaning
819
820   ### Build
821
822   Deterministic normalization:

```

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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
818 ## Phase 7 - Cleaning
819
820 ### Build
821
822 Deterministic normalization:
823
824 - line ending normalization
825 - safe whitespace normalization
826 - extraction artifact removal
827 - preserve identifiers
828 - preserve labels
829 - preserve business punctuation
830 - preserve table meaning
831
832 Do not rewrite business meaning.
833
834 ### Gate
835
836 Raw extraction and cleaned extraction can be compared for representative fixtures.
837
838 ---
839
840 ## Phase 8 - Rule-based structured parsing
841
842 ### Build
843
844 ```text
845 RequirementParser
846 AcceptanceCriteriaParser
847 TestCaseParser
848 DefectParser
849 ```
850
851 Priority:
852
853 ```text
854 explicit labels
855
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS Show desktop
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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
850
851 Priority:
852
853 ```text
854 explicit labels
855 -> headings
856 -> table structure
857 -> IDs
858 -> deterministic patterns
859 ```
860
861 Example:
862
863 ```text
864 TC-231 | Address validation | Expected: AddressLine2 is optional
865 ```
866
867 may produce fields directly supported by that source.
868
869 Do not generate a root cause or expected result that does not exist in the source.
870
871 ### Gate
872
873 Known fixture values match expected parser output.
874
875 ---
876
877 ## Phase 9 - Groq fallback parser
878
879 ### Build
880
881 Use Groq only when deterministic parsing cannot reliably structure the source.
882
883 ### Contract
884
885 Require schema-valid structured output containing evidence.
886
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
```

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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
876
877 ## Phase 9 - Groq fallback parser
878
879 ### Build
880
881 Use Groq only when deterministic parsing cannot reliably structure the source.
882
883 ### Contract
884
885 Require schema-valid structured output containing evidence.
886
887 Conceptually:
888
889 ```json
890 {
891   "field": "expectedResult",
892   "value": "...",
893   "evidence": "...",
894   "evidenceStatus": "SUPPORTED"
895 }
896 ```
897
898 Unsupported output becomes:
899
900 ```text
901 INSUFFICIENT_EVIDENCE
902 ```
903
904 ### Gate
905
906 - malformed Groq response is rejected
907 - unsupported field is not persisted as fact
908 - source evidence is retained
909 - parser method is recorded
910
911 ---
912
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
```

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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
913 ## Phase 10 - Structure-aware chunking
914
915 ### Build
916
917 Prefer:
918
919 ```text
920 section
921 -> paragraph
922 -> table/record
923 -> configured size limit
924 -> configured overlap
925 ```
926
927 Keep Requirement/Test Case/Defect records together where possible.
928
929 All chunk limits and overlap values come from environment configuration.
930
931 ### Gate
932
933 Inspect representative chunks manually and confirm:
934
935 - no unexplained content loss
936 - meaningful boundaries
937 - source location retained
938 - entity association retained
939
940 ---
941
942 ## Phase 11 - Embedding service
943
944 ### Build
945
946 ```text
947 EmbeddingService
948 +-- MistralEmbeddingProvider
949 +-- OpenAIEmbeddingProvider
950 ```
951
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
942 ## Phase 11 -- Embedding service
943
944 ### Build
945
946 ```text
947 EmbeddingService
948   +-- MistralEmbeddingProvider
949   +-- OpenAIEmbeddingProvider
950   ...
951
952 Selected behavior:
953
954 ```text
955 Mistral
956 |
957 +-- success -> use vector
958 |
959 +-- failure -> OpenAI fallback
960 ...
961
962 Persist:
963
964 ```text
965 provider
966 model
967 dimension
968 createdAt
969 sourceChunkId
970 ...
971
972 Mistral's official API documents the embeddings endpoint, model selection, text input, and output-dimension option where supported. [Mistral Embedding
973
974 ### Critical verification
975
976 The MongoDB Vector Search index dimension must match the vector dimension used by the stored embeddings. MongoDB explicitly documents that the embeddi
977
978 ---
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
980 ## Phase 12 -- MongoDB persistence
981
982 ### Repositories
983
984 ```text
985 ProjectRepository
986 DocumentRepository
987 DocumentVersionRepository
988 IngestionJobRepository
989 ChunkRepository
990 RequirementRepository
991 AcceptanceCriteriaRepository
992 TestCaseRepository
993 DefectRepository
994 RelationshipRepository
995 ProvenanceRepository
996 VerificationRepository
997 ...
998
999 ### Persistence sequence
1000
1001 ```text
1002 document/version
1003 -> extraction/provenance
1004 -> entities
1005 -> relationships
1006 -> chunks
1007 -> embeddings
1008 -> verification
1009 ...
1010
1011 Use transactions only where the actual consistency requirement justifies them; do not assume separate MongoDB writes are automatically atomic.
1012
1013 ### Gate
1014
1015 Read back every expected record from MongoDB.
1016
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
1019 ## Phase 13 -- MongoDB Vector Search
1020
1021 ### Build
1022
1023 Create a vector index for the chunk embedding field.
1024
1025 Include project filtering fields in the vector-search design.
1026
1027 MongoDB documents that Vector Search supports semantic retrieval and pre-filtering on indexed fields, making project-scoped retrieval feasible. [Mong
1028
1029 ### Gate
1030
1031 Run a known semantic smoke query against an ingested fixture and verify that the expected source chunk is retrievable.
1032
1033 ---
1034
1035 ## Phase 14 -- Evidence-backed relationships
1036
1037 Supported relationship types initially:
1038
1039 ```text
1040 Requirement -> AcceptanceCriteria
1041 Requirement -> TestCase
1042 Defect -> Requirement
1043 Defect -> TestCase
1044 ```
1045
1046 Only persist a documented relationship when the source provides evidence.
1047
1048 Do not convert semantic similarity into a factual relationship.
1049
1050 ---
1051
1052 ## Phase 15 -- Verification engine
1053
1054 ### Checks
1055
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
1052 ## Phase 15 -- Verification engine
1053
1054 ### Checks
1055
1056 1. Document exists.
1057 2. Version exists.
1058 3. Content hash matches.
1059 4. Extraction completed.
1060 5. Extracted content exists.
1061 6. Entities satisfy schemas.
1062 7. Entity evidence exists.
1063 8. Chunks exist and preserve provenance.
1064 9. Embeddings exist.
1065 10. Provider/model/dimension are recorded.
1066 11. MongoDB records can be read back.
1067 12. Retrieval smoke test returns the newly ingested source.
1068
1069 ### Completion rule
1070
1071 ```text
1072 All required checks pass
1073 |
1074 v
1075 INGESTION_COMPLETE
1076 ```
1077
1078 Otherwise:
1079
1080 ```text
1081 PARTIALLY_PROCESSED
1082 or
1083 FAILED
1084 ```
1085
1086 ---
1087
1088 ## Phase 16 -- Status APIs
1089
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
1088 ## Phase 16 -- Status APIs
1089
1090 ```text
1091 GET /api/projects/:projectId/documents
1092 GET /api/projects/:projectId/documents/:documentId
1093 GET /api/projects/:projectId/ingestion-jobs/:jobId
1094 GET /api/projects/:projectId/documents/:documentId/verification
1095 ```
1096
1097 Expose:
1098
1099 - current status
1100 - current stage
1101 - progress
1102 - version
1103 - chunk count
1104 - entity counts
1105 - embedding status
1106 - verification status
1107 - errors
1108 - timestamps
1109
1110 Never expose credentials.
1111
1112 ---
1113 ## Phase 17 -- React UI
1114
1115 Create:
1116
1117 ```text
1118 apps/web/src/features/ingestion/
1119 UploadPage
1120 UploadForm
1121 IngestionStatus
1122 VerificationSummary
1123 ErrorSummary
1124 ```
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
1113 ## Phase 17 -- React UI
1114
1115 Create:
1116
1117 ```text
1118 apps/web/src/features/ingestion/
1119 UploadPage
1120 UploadForm
1121 IngestionStatus
1122 VerificationSummary
1123 ErrorSummary
1124 ```
1125
1126 Journey:
1127
1128 ```text
1129 Project
1130 -> File
1131 -> Upload
1132 -> Job
1133 -> Processing
1134 -> Extraction
1135 -> Entities/chunks
1136 -> Embeddings
1137 -> Verification
1138 -> INGESTION_COMPLETE
1139 ```
1140
1141
1142 ### Gate
1143
1144 A developer can visually prove the backend ingestion state without inspecting MongoDB directly.
1145
1146 ---
1147 ## Phase 18 -- Logging + errors
1148
1149
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
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D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
1148 ## Phase 18 -- Logging + errors
1149
1150 Every ingestion log should include, where available:
1151
1152 ```text
1153 requestId
1154 jobId
1155 projectId
1156 documentId
1157 documentVersionId
1158 stage
1159 errorCode
1160 duration
1161 ```
1162
1163 Error categories:
1164
1165 ```text
1166 VALIDATION_ERROR
1167 STORAGE_ERROR
1168 EXTRACTION_ERROR
1169 PARSING_ERROR
1170 LLM_ERROR
1171 EMBEDDING_ERROR
1172 DATABASE_ERROR
1173 QUEUE_ERROR
1174 VERIFICATION_ERROR
1175 ```
1176
1177 Never log API keys, access tokens, connection strings, or unnecessary sensitive source content.
1178
1179 ---
1180 ## Phase 19 -- Testing
1181
1182 ### Unit tests
1183
1184
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
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D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
1181 ## Phase 19 -- Testing
1182
1183 ### Unit tests
1184
1185 - environment validation
1186 - file validation
1187 - hashing
1188 - regex utilities
1189 - cleaning
1190 - parsers
1191 - chunking
1192 - provenance
1193 - schemas
1194 - embedding provider selection
1195 - retry classification
1196
1197 ### Integration tests
1198
1199 ```text
1200 upload -> storage
1201 upload -> queue
1202 worker -> extractor
1203 worker -> parser
1204 worker -> chunker
1205 worker -> embedding
1206 worker -> MongoDB
1207 worker -> verification
1208 ```
1209
1210 ### Fixtures
1211
1212 ```text
1213 tests/fixtures/
1214 requirement.pdf
1215 testcase.docx
1216 defects.xlsx
1217 notes.txt
1218 ```
1219
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
```

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D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
1210 ### Fixtures
1211
1212 ```text
1213 tests/fixtures/
1214   requirement.pdf
1215   testcase.docx
1216   defects.xlsx
1217   notes.txt
1218 ```
1219
1220 Fixture information must be explicitly created test data, not fabricated production evidence.
1221
1222 ### End-to-end acceptance path
1223
1224 ```text
1225 UPLOAD
1226 -> VALIDATE
1227 -> QUEUE
1228 -> EXTRACT
1229 -> PARSE
1230 -> CHUNK
1231 -> EMBED
1232 -> STORE
1233 -> VERIFY
1234 -> RETRIEVE
1235 -> INGESTION_COMPLETE
1236 ```
1237
1238 ---
1239
1240 # 10. Final ingestion readiness checklist
1241
1242 ## Configuration
1243
1244 - // All environment-dependent values come from `.env`.
1245 - // `example.env` exists.
1246 - // No secret is hard-coded.
```

User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS

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D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
1240 # 10. Final ingestion readiness checklist
1241
1242 ## Configuration
1243
1244 - // All environment-dependent values come from `.env`.
1245 - // `example.env` exists.
1246 - // No secret is hard-coded.
1247 - // Startup validates configuration.
1248
1249 ## Upload
1250
1251 - // PDF works.
1252 - // DOCX works.
1253 - // XLSX works.
1254 - // TXT works.
1255 - // Unsupported files are rejected.
1256 - // Size validation works.
1257 - // MIME/content validation works.
1258 - // Exact duplicates are detected.
1259 - // Changed documents create versions.
1260
1261 ## Storage
1262
1263 - // Local storage works in development.
1264 - // S3 adapter works for deployment.
1265 - // Original bytes can be recovered.
1266 - // Storage metadata is persisted.
1267
1268 ## Queue
1269
1270 - // BullMQ job is created.
1271 - // Worker consumes the job.
1272 - // Retry works for transient failures.
1273 - // Failed jobs are visible.
1274 - // Concurrency is configurable.
1275
1276 ## Extraction
```

User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS

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D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
1276 ## Extraction
1277 - // PDF extraction works.
1278 - // DOCX extraction works.
1279 - // XLSX extraction works.
1280 - // TXT extraction works.
1281 - // Source locations are preserved where available.
1282
1283 ## Parsing
1284 - // Requirement parser works.
1285 - // Acceptance Criteria parser works.
1286 - // Test Case parser works.
1287 - // Defect parser works.
1288 - // Rule parser runs before Groq.
1289 - // Groq response is schema validated.
1290 - // Unsupported values become `Insufficient evidence`.
1291
1292 ## Chunking
1293 - // Structure-aware chunking works.
1294 - // Limits come from configuration.
1295 - // Provenance survives chunking.
1296
1297 ## Embeddings
1298 - // Mistral works.
1299 - // OpenAI fallback works.
1300 - // Provider/model/dimension are recorded.
1301 - // MongoDB vector index dimension matches embeddings.
1302
1303 ## MongoDB
1304 - // Documents persist.
1305 - // Versions persist.
1306 - // Entities persist.
1307 - // Evidence-backed relationships persist.
1308
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User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
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D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
1307 ## MongoDB
1308 - // Documents persist.
1309 - // Versions persist.
1310 - // Entities persist.
1311 - // Evidence-backed relationships persist.
1312 - // Chunks persist.
1313 - // Embeddings persist.
1314 - // Provenance persists.
1315 - // Verification results persist.
1316
1317 ## Verification
1318 - // Storage verification works.
1319 - // Extraction verification works.
1320 - // Entity verification works.
1321 - // Chunk verification works.
1322 - // Embedding verification works.
1323 - // Retrieval smoke test works.
1324 - // Only verified jobs become `INGESTION_COMPLETE`.
1325
1326 ## UI
1327 - // Upload works.
1328 - // Status is visible.
1329 - // Current stage is visible.
1330 - // Counts are visible.
1331 - // Verification result is visible.
1332 - // Errors are visible.
1333
1334 ---
1335
1336 # 11. Deliberately outside the first-version ingestion scope
1337 These are future adapters, not assumptions for Version 1:
1338 - Jira
1339
1340
1341
1342
1343
User Defined language file - Markdown (preinstalled) length: 26,147 lines: 1,437 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS
```

```
# 11. Deliberately outside the first-version ingestion scope

These are future adapters, not assumptions for Version 1:

- Jira
- ADO
- TestRail
- Zephyr
- Xray
- Figma
- Swagger/API live ingestion
- GitHub
- Bitbucket
- meeting recordings
- Confluence
- wiki connectors
- external defect databases
- release-note systems

Developer repositories remain non-vector according to the stated requirement and are not part of the first uploaded-file ingestion implementation.

---

# 12. Implementation order

```text
Phase 0 Bootstrap
Phase 1 MongoDB + Redis
Phase 2 Project boundary
Phase 3 File storage
Phase 4 Upload + validation
Phase 5 BullMQ
Phase 6 Extraction
Phase 7 Cleaning
Phase 8 Rule parsing
Phase 9 Groq fallback
Phase 10 Chunking
```

```
D:\TestLeaf\Program\August\phase2\RAG module_12thSept_Week7_Week8_Week9\26th Sept-AgentBuilding\AI_QA_Knowledge_Assistant_Ingestion_Architecture.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
AI_QA_Knowledge_Assistant_Ingestion_Architecture.md
1392 # 13. Technology evidence
1393
1394 - Mistral Embeddings API: https://docs.mistral.ai/api/endpoint/embeddings
1395 - MongoDB Vector Search: https://www.mongodb.com/docs/vector-search/
1396 - MongoDB `VectorSearch`: https://www.mongodb.com/docs/manual/reference/operator/aggregation/vectorsearch/
1397 - BullMQ Workers: https://docs.bullmq.io/guide/workers
1398 - BullMQ retries: https://docs.bullmq.io/guide/retrying-failing-jobs
1399 - AWS SDK for JavaScript v3: https://docs.aws.amazon.com/sdk-for-javascript/v3/developer-guide/getting-started-nodejs.html
1400
1401 These sources establish technology capabilities only; they are not business requirements.
1402
1403 ---
1404 # 14. Final architecture rule
1405
1406 ```text
1407
1408 SOURCE EVIDENCE
1409 |
1410 v
1411 INGESTION
1412 |
1413 +-----+
1414 | EXTRACT | PARSE | CHUNK |
1415 +-----+
1416 |
1417 +-----+
1418 | EMBEDDING |
1419 |
1420 v
1421 MONGODB
1422 |
1423 VERIFICATION
1424 |
1425 +-----+
1426 | FAIL/PARTIAL | COMPLETE |
1427 +-----+
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